

## BINAY—Lighting Up The Future

## Binay LED Aviation Obstruction Lights



**BINAY LED-based MEDIUM INTENSITY  
Aviation Obstruction Light Beacon**

As per the International Civil Aviation Organisation (ICAO) requirements for use on structures above 45m

Available in Low Intensity and Medium Intensity versions (as per the International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with a 5-year warranty

### LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave, and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED intensity degradation over its operating lifetime



**THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED  
PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT**



**binay opto electronics private ltd.**

44, Armenian Street, Calcutta 700 001, India.  
Telephone : (033) 2242 9082, 2210 3807, 2210 2039  
email : info@binayLED.com, binay@vsnl.com,  
Fax : 91-33-2242 1493 <http://www.binayLED.com>

fectively means positively charged (p-type) and negatively charged (n-type) conduction is possible in a single oxide material. The researchers claim it to be the first functional oxide thin film that can do so.

When more than one oxide is used at a p-n junction (one positively doped and the other negatively doped), performance suffers due to mismatches. Using the same oxide eliminates the problems caused at such a p-n junction. This development could pave the way for new, high-power semiconductor devices, ultra high-density device features and smart sensors.

The team used lasers to create positively charged nickel-oxide (NiO) thin films, then converted the top layer of those films into n-type. The ability to control the thickness of the n-layer allowed the researchers to control the depth and characteristics of the p-n junction. This selectivity in space and time provided them control to 'write' p-n junctions by laser beams, in a way that has never been done before.

## Unusual electrons created using molecular graphene

Scientists at Stanford University have managed to recreate graphene from scratch, paving the way for the ability to tweak existing materials according to our needs. Graphene is a one-atom-thick planar sheet of sp<sup>2</sup>-bonded carbon atoms, densely packed in a honeycomb crystal lattice.

The physicists, led by Hari Manoharan, created what they term 'molecular graphene' by laying out individual carbon-monoxide molecules on a perfectly smooth sheet of copper. The substance isn't graphene, but acts like it. The carbon-monoxide molecules that aren't part of the substance block the free-flowing electrons on the copper sheet, forcing them into a honeycomb pattern resembling graphene. After recreating graphene, the team tweaked it by repositioning carbon monoxide carefully to change the electron flow. By carefully working out where the tweaks needed to be applied, the team managed to simulate a magnetic field of 60 tesla—30 per cent greater than the greatest continuous magnetic field ever created.

The ability to make designer structures from electrons opens up possibilities for identifying new nanoscale materials with useful electronic properties.

## Diamond thin films could solve thermal bottleneck

For decades, engineers have sought to build more efficient electronic devices by reducing the size of their components. In the process, however, they have reached a thermal bottleneck.

The unusually attractive thermal properties of diamond thin films have led scientists to suggest using this material as a heat-sink that could be integrated with a number of different semiconducting materials. While properties of diamond thin films are relatively well-understood, two new studies performed at the US Department of Energy's Argonne National Laboratory have revealed a new pathway for ma-